Experiment 7

Non-Preemptive SJF Scheduling

code:

```
#include <stdio.h>

int main(void) {
   int n, i, time = 0, done = 0;
   int at[20], bt[20], wt[20], tat[20], used[20] = {0};
   float sw = 0, st = 0;

   printf("Number of processes (max 20): ");
   scanf("%d", &n);

   if (n < 1 || n > 20)
      return 1;

   for (i = 0; i < n; i++) {
      printf("P%d arrival and burst: ", i + 1);
      scanf("%d%d", &at[i], &bt[i]);
   }

   printf("\nProcess\tWT\tTAT\n");

   while (done < n) {
      int k = -1;

      for (i = 0; i < n; i++) {
         if (!used[i] && at[i] <= time &&
            (k == -1 || bt[i] < bt[k]))
            k = i;
      }
```

```
        if (k == -1) {
            time++;
            continue;
        }

        wt[k] = time - at[k];
        time += bt[k];
        tat[k] = time - at[k];

        used[k] = 1;
        done++;

        sw += wt[k];
        st += tat[k];

        printf("P%d\t%d\t%d\n", k + 1, wt[k], tat[k]);
    }

    printf("Average waiting time = %.2f\n", sw / n);
    printf("Average turnaround time = %.2f\n", st / n);

    return 0;
}
```

Output:


Run
Debug
Stop
Share
Save
{ } Beautify
main.c
if (!used[i] && at[i] <= time &&
(k == -1 || bt[i] < bt[k]))
k = i;
}
if (k == -1) {
time++;
continue;
}
wt[k] = time - at[k];
time += bt[k];
tat[k] = time - at[k];
used[k] = 1;
input
Number of processes (max 20): 3
P1 arrival and burst: 0 5
P2 arrival and burst: 3 4
P3 arrival and burst: 5 7
Process WT TAT
P1 0 5
P2 2 6
P3 4 11
Average waiting time = 2.00
Average turnaround time = 7.33
...Program finished with exit code 0
Press ENTER to exit console.